A Counterexample to the Finite-Complexity-Rate Inductive-Limit Question

arXiv automatic math run `fa_banach_001`

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Status

This packet gives a candidate counterexample, likely valid, to the inductive-limit question in Tian–Yu [4]. The result is stronger than the earlier partial packet: it gives a negative answer both for injective/direct-union inductive limits and for filtered colimits with arbitrary bonding homomorphisms.

The supplied report

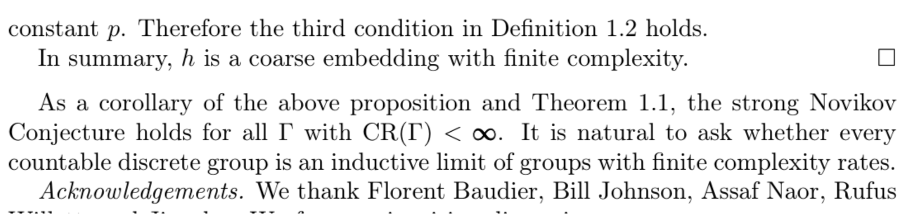
`finite_complexity_rate_counterexample.tex/pdf`

has been copied into the evidence folder. The previous direct-union reduction packet has been absorbed into the history folder of this packet.

Source Question

The source is Geng Tian and Guoliang Yu, *Embedding complexity into Banach spaces and the strong Novikov conjecture*, arXiv:2605.12930v1. After Definition 1.4 and Proposition 1.5, the paper asks on page 6:

> "It is natural to ask whether every countable discrete group is an inductive limit of groups with finite complexity rates."


constant p. Therefore the third condition in Definition 1.2 holds.

In summary, h is a coarse embedding with finite complexity. □

As a corollary of the above proposition and Theorem 1.1, the strong Novikov Conjecture holds for all Γ with $\mathrm{CR}(\Gamma) < \infty$. It is natural to ask whether every countable discrete group is an inductive limit of groups with finite complexity rates.

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Definitions Used

Let Γ be a countable group with a left-invariant proper metric. In Definition 1.4 of [4], finite complexity rate means $\mathrm{CR}(\Gamma) < \infty$, equivalently $\mathrm{DR}(\Gamma) > 1$. Thus there is an element

$$\mathbf{a} = (a_0, a_1, \ldots, a_k) > 1$$

in the lexicographic order used in the source, a constant $C > 0$, thresholds $N_p > 0$, and maps

$$\phi_p : \Gamma \longrightarrow S(\ell^{2p}(\mathbb{N}))$$

such that

$$d(x,y) \le 1 \Longrightarrow \|\phi_p(x) - \phi_p(y)\|_{2p} \le \varepsilon_p, \tag{1}$$
$$d(x,y) \ge N_p \Longrightarrow \|\phi_p(x) - \phi_p(y)\|_{2p} \ge 1, \tag{2}$$

where

$$\varepsilon_p = \frac{C}{p^{a_0}(\log p)^{a_1} \cdots (\log_k p)^{a_k}}.$$

The source's truncated convention for small iterated logarithms is irrelevant for the asymptotic argument. Since $\mathbf{a} > 1$, one has

$$p\varepsilon_p \longrightarrow 0. \tag{1}$$

Proof Intuition

Finite complexity rate would provide almost unit-scale-constant maps ϕ_p into the unit spheres of ℓ^{2p}, with variations $\varepsilon_p = o(1/p)$, while still separating sufficiently far group elements. Expanders forbid this rate. If an expander family coarsely embeds in a finitely generated group, then sufficiently many pairs in a large graph are sent beyond the separation threshold. The L_q Poincare inequality for expanders, with $q = 2p$, forces the edge variation of the composed map to be at least a constant multiple of $1/p$. This contradicts $p\varepsilon_p \to 0$. Osajda supplies a finitely presented group with exactly this coarse expander containment, giving the counterexample.

Direct-Union Reduction

Lemma 1 (Subgroup heredity)**.** *If a countable group Γ has finite complexity rate, then every subgroup $\Lambda \le \Gamma$ has finite complexity rate.*

Proof. Restrict a left-invariant proper metric on Γ to Λ. It is again left-invariant and proper. Restricting the maps ϕ_p preserves the variation and separation estimates, and [4] states that the definition is independent of the chosen left-invariant proper metric. □

Proposition 2 (Directed-union reduction)**.** *For a countable group Γ, the following are equivalent:*

Γ is a directed union of finite-complexity-rate subgroups

and

every finitely generated subgroup of Γ has finite complexity rate.

Proof. If $\Gamma = \bigcup_{i \in I} \Gamma_i$, with I directed and all Γ_i of finite complexity rate, then any finitely generated subgroup $H \le \Gamma$ has a finite generating set contained in one stage Γ_i. Lemma 1 gives finite complexity rate for H.

Conversely, the finitely generated subgroups of Γ, ordered by inclusion, form a directed system whose union is Γ. If each of them has finite complexity rate, this is the required directed union. □

The Expander Obstruction

Let $X = (V, E)$ be a finite connected d-regular graph, and let A_X be the normalized adjacency operator. A sequence (X_n) is a spectral expander sequence with gap $\sigma > 0$ if $|V(X_n)| \to \infty$ and

$$1 - \lambda_2(A_{X_n}) \geq \sigma$$

for every n.

We say that (X_n) coarsely embeds into a finitely generated group Γ if there are maps $\iota_n : V(X_n) \to \Gamma$ and nondecreasing functions $\rho_-, \rho_+ : [0, \infty) \to [0, \infty)$, independent of n, with $\rho_-(t) \to \infty$, such that

$$\rho_-\big(d_{X_n}(u, v)\big) \leq d_\Gamma\big(\iota_n(u), \iota_n(v)\big) \leq \rho_+\big(d_{X_n}(u, v)\big).$$

Lemma 3 (L_q Poincare inequality). *Let $X = (V, E)$ be a finite d-regular graph whose normalized adjacency operator has spectral gap at least $\sigma > 0$. For every $q \geq 2$ and every map $f : V \to \ell^q$,*

$$\left(\frac{1}{|V|^2} \sum_{u,v \in V} \|f(u) - f(v)\|_q^q\right)^{1/q} \leq \frac{2q}{\sqrt{\sigma}} \left(\frac{1}{d|V|} \sum_{u \in V} \sum_{v \sim u} \|f(u) - f(v)\|_q^q\right)^{1/q}.$$

Proof. This is the L_q expander Poincare inequality of Matousek, in the form quoted as Corollary 4.5 in Naor–Silberman [2]. □

Theorem 4 (Expanders force the critical $1/p$ rate). *Let Γ be finitely generated. Suppose a bounded-degree spectral expander sequence with gap $\sigma > 0$ coarsely embeds into Γ. If maps $\phi_p : \Gamma \to S(\ell^{2p}(\mathbb{N}))$ satisfy* (1) *and* (2)*, then there is a constant $c > 0$, independent of p, such that*

$$p\varepsilon_p \geq c \qquad (p \in \mathbb{N}).$$

Consequently Γ has infinite complexity rate.

Proof. Use a word metric on Γ, and set

$$L = \max\{1, \lceil \rho_+(1) \rceil\}.$$

Since the word metric is geodesic, (1) implies

$$d_\Gamma(x, y) \leq L \Longrightarrow \|\phi_p(x) - \phi_p(y)\|_{2p} \leq L\varepsilon_p. \tag{2}$$

Fix p, set $q = 2p$, and choose R_p with $\rho_-(R_p) \geq N_p$. Since the degrees are bounded, every radius $(R_p - 1)$-ball in the graphs has size at most a number B_p independent of n. Choose n with $|V(X_n)| \geq 2B_p$. Then at least half of the ordered pairs $(u, v) \in V(X_n)^2$ satisfy $d_{X_n}(u, v) \geq R_p$.

Let $f = \phi_p \circ \iota_n$. For those far pairs, the coarse lower control and (2) give $\|f(u) - f(v)\|_q \geq 1$. Hence

$$\left(\frac{1}{|V(X_n)|^2} \sum_{u,v \in V(X_n)} \|f(u) - f(v)\|_q^q\right)^{1/q} \geq 2^{-1/q}. \tag{3}$$

For adjacent vertices $u \sim v$, coarse upper control gives $d_\Gamma(\iota_n(u), \iota_n(v)) \leq \rho_+(1) \leq L$, so by (2)

$$\|f(u) - f(v)\|_q \leq L\varepsilon_p.$$

Applying Lemma 3 and (3),

$$2^{-1/q} \leq \frac{2q}{\sqrt{\sigma}} L\varepsilon_p.$$

With $q = 2p$, this gives

$$p\varepsilon_p \geq \frac{\sqrt{\sigma}}{4L} 2^{-1/(2p)} \geq \frac{\sqrt{\sigma}}{4\sqrt{2}\,L}.$$

This lower bound contradicts $p\varepsilon_p \to 0$ whenever Γ has finite complexity rate. $\square$

The Counterexample

Osajda constructed finitely generated groups whose Cayley graphs contain isometrically embedded expander sequences [3, Theorem 3.2 and Corollary 3.3]. More strongly for the present purpose, he obtained closed aspherical manifolds of dimension at least four whose fundamental groups contain coarsely embedded expanders [3, Corollary 3.5]. Such a fundamental group is finitely presented.

Corollary 5. *There exists a countable finitely presented group G with* $\mathrm{CR}(G) = \infty$*.*

Proof. Take G to be one of Osajda's finitely presented fundamental groups coarsely containing a bounded-degree expander family. Cheeger's inequality turns the uniform combinatorial expansion in the construction into a uniform spectral gap after passing to a fixed-degree regular expander family. Theorem 4 gives $\mathrm{CR}(G) = \infty$. $\square$

Lemma 6 (Finitely presented colimits retract from a stage)**.** *Let $G \cong \varinjlim_i G_i$ be a filtered colimit of groups. If G is finitely presented, then for some i there is a homomorphism $s : G \to G_i$ such that the canonical map $\lambda_i : G_i \to G$ satisfies $\lambda_i \circ s = \mathrm{id}_G$. In particular, G is isomorphic to a subgroup of G_i.*

Proof. Write $G = \langle x_1, \ldots, x_m \mid r_1, \ldots, r_k \rangle$. Choose representatives of the finitely many generators in stages of the filtered system and pass to one common later stage. Each relator maps to the identity in the colimit, so after passing to a further common later stage, all relators are already trivial. The chosen representatives therefore define a homomorphism $s : G \to G_i$, and $\lambda_i \circ s$ fixes the generators of the presented group G. Thus $\lambda_i \circ s = \mathrm{id}_G$, so s is injective. $\square$

Theorem 7 (Negative answer)**.** *The inductive-limit question in [4] has a negative answer:*

1. *there is a countable finitely presented group that is not a directed union of groups with finite complexity rates;*

2. *the same group is not a filtered colimit of groups with finite complexity rates, even when the bonding homomorphisms are arbitrary and need not be injective.*

Proof. Let G be the finitely presented group from Corollary 5.

If G were a directed union of finite-complexity-rate subgroups, its finite generating set would lie in one stage. That stage would contain all of G, so G itself would have finite complexity rate, contradicting $\mathrm{CR}(G) = \infty$.

Now suppose $G \cong \varinjlim_i G_i$, with all G_i of finite complexity rate. By Lemma 6, G embeds in one G_i. Lemma 1 then implies finite complexity rate for G, again a contradiction. $\square$

Verification Notes

The proof is noncomputational. The quantitative point to check is the dependence on q in Lemma 3; the needed bound is linear in q, so with $q = 2p$ it gives the lower bound $p\varepsilon_p \geq c$. The group theoretic steps are formal once finite presentability and coarse expander containment are available.

Novelty and Literature Status

The exact notion of finite complexity rate was introduced in arXiv:2605.12930v1, submitted May 13, 2026. The previous run packet searched on June 16, 2026 for the arXiv id, the paper title together with "inductive limit", and the phrase "finite complexity rates inductive limit countable discrete group"; no later answer to the exact question was found. The supplied report records another bounded search by title, arXiv id, the phrase "finite complexity rate", and the inductive-limit question as of June 21, 2026, again without locating a later explicit answer. During this upgrade, the run indexes and web searches for the arXiv id, finite complexity rate, and the supplied report title found no duplicate packet or published answer.

This packet therefore records the result as a candidate counterexample produced by synthesis of known ingredients: Matousek/Naor–Silberman expander Poincare inequalities and Osajda's groups containing expanders. It makes no priority claim beyond that synthesis.

Human Review Recommendation

Review as a candidate full negative answer to the source question. The main checks are:

1. the source Definition 1.4 really implies $p\varepsilon_p \to 0$ for finite complexity rate;
2. the cited Poincare inequality has the required linear q-dependence for ℓ^q;
3. Osajda's Corollary 3.5 supplies a finitely presented group coarsely containing a bounded-degree expander sequence in the sense used above.

References

[1] Jiri Matousek, *On embedding expanders into ℓ_p spaces*, Israel Journal of Mathematics **102** (1997), 189–197.

[2] Assaf Naor and Lior Silberman, *Poincare inequalities, embeddings, and wild groups*, Compositio Mathematica **147** (2011), no. 5, 1546–1572, arXiv:1005.4084.

[3] Damian Osajda, *Small cancellation labellings of some infinite graphs and applications*, Acta Mathematica **225** (2020), no. 1, 159–191, arXiv:1406.5015.

[4] Geng Tian and Guoliang Yu, *Embedding complexity into Banach spaces and the strong Novikov conjecture*, arXiv:2605.12930v1, 2026.